Patient Length of Stay (LoS) Prediction

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Introduction

In 2024, there were 35.6 million hospital admissions in the United States (AHA), with an average length of stay (LoS) of 6 days (OECD). Prolonged LoS negatively affects both patients and hospital staff, leading to avoidable health complications and increased stress, respectively (Rojas-García *et al.*, 2018). The ability to accurately predict LoS could enable more effective resource planning and enhanced operational efficiency, improving inpatient outcomes while relieving pressure on healthcare workers. The goal of this project is to develop a predictive model to determine LoS using only admission-level data, enabling better resource allocation and patient care before a patient's hospital course even begins. To allow the data to be manageable, we limit the scope of the project to predicting lengths of stay in Texas; this could be generalized to national data in future work.

Stakeholders and KPIs

Our primary stakeholders – patients, hospital systems and administrators, and clinical teams – each have distinct but overlapping interests in LoS prediction. Patients benefit from more transparent care timelines and reduced risks of complications tied to prolonged stays. Hospital administrators require actionable predictions to optimize staffing, bed allocation, and discharge planning. Clinical teams need reliable estimates to effectively coordinate patient care. Given these varied needs, we evaluated models using a combination of metrics: Root Mean Squared Error, Mean Absolute Error, Huber Loss, and Quantile Loss. A model was considered clinically and operationally viable only if it achieved meaningful improvement relative to the baseline.

Dataset and Preparation

The dataset consists of over 7 million hospitalizations within Texas from 2017-2019, spanning more than 160 features. To reflect a realistic prediction scenario, we restricted our feature set to admission-only variables: patient demographics, diagnosis codes present at admission and facility and location information. We further added an urban/rural flag and the calculated distance between each patient's residence ZIP code and the hospital location, capturing healthcare access patterns that may influence patient care.

Models and Results

A linear regression model was established as the baseline to benchmark the performance of more complex approaches. While interpretable and computationally efficient, the linear baseline was limited in its ability to capture the nonlinearity inherent to hospital admission data, yielding insufficient prediction accuracy. Four tree-based models were subsequently developed and compared: Random Forest, XGBoost, LightGBM, and Catboost. RandomForestRegressor leverages an ensemble of decorrelated decision trees to reduce variance and improve generalization. XGBoost applies gradient boosting with regularization to iteratively correct prediction errors, offering strong performance on structured tabular data. LightGBM is a highly efficient gradient boosting framework that uses histogram-based algorithms and leaf-wise growth, enabling faster training on large datasets while maintaining competitive accuracy. Catboost, also a gradient boosting framework, is specifically designed to handle categorical features natively without extensive preprocessing – a key advantage given categorical variables prevalent in our dataset.

CatBoost was selected as the final model, achieving a RMSE of 10.361, an MSE of 3.122, and a Quantile Loss of 1.606, respectively. These results represent a substantial improvement over the linear baseline across all KPIs, demonstrating strong predictive accuracy and robust performance across the full LoS distribution.

Future Steps

Future work will focus on subgroup analysis by diagnosis category, patient demographics, and facility type. This may reveal disparities in levels of care that warrant targeted model refinement. Finally, validation in live hospital environments is essential to assess real-world performance and inform integration into healthcare system workflows.